

1. Executive Summary

		Traffic Summary		Count	
		Total Flows Observed		14526	
		Total Packets Analyzed		2701090103	
		Malicious Packets		2697949853	
		True Negatives Passed		11154	
		Attack Vector		Count	
		ICMP Flood		0	
		SYN Flood		0	
		UDP Flood		11	
		Uncertain		3	
		Action		Count	
		Quarantined		3	
		Time Ban		3	
		Blackhole		0	
		Released		1	

Metric	Value
Total Threats Mitigated	14
High Priority	10
Low Priority	4
Manual Releases	0
Manual Blocks	0
ML Metrics	Value
False Positives	1
FP Rate	0.01%
IF Precision	99.97%
IF Recall	95.52%
IF F1-Score	97.70%
IF Accuracy	98.91%

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.97%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.52%	Share of actual attacks successfully flagged
F1-Score	97.70%	Balanced measure combining Precision and Recall
Accuracy	98.91%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.01%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.48%	Actual attacks that went undetected
True Positive Rate (TPR)	95.52%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.99%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 3371	FN 158
Actual: Normal	FP 1	TN 10996

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	3371

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.97%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.52%	Share of actual attacks successfully flagged
F1-Score	97.70%	Balanced measure combining Precision and Recall
Accuracy	98.91%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.01%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.48%	Actual attacks that went undetected
True Positive Rate (TPR)	95.52%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.99%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 3371	FN 158
Actual: Normal	FP 1	TN 10996

2d. Response Latency

Metric	Value	Description
Detection Time	154.1 ms	Average time taken to detect an attack
Mitigation Response Time	0.0 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	0.94%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	2.52%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	6.01%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.20	2	2	2026-09-15 19:56:23	2026-09-15 19:57:34
10.0.0.21	2	2	2026-09-15 19:56:24	2026-09-15 19:57:35
10.0.0.22	2	2	2026-09-15 19:56:26	2026-09-15 19:57:37
10.0.0.8	1	1	2026-09-15 19:57:23	2026-09-15 19:58:24

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-15 19:56:23	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 19:56:24	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 19:56:26	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 19:56:34	10.0.0.20	DDoS	UDP Flood	75.6%	High	Time Ban (1m)
2026-09-15 19:56:34	10.0.0.20	DDoS	UDP Flood	75.6%	High	Time Ban
2026-09-15 19:56:35	10.0.0.21	DDoS	UDP Flood	75.3%	High	Time Ban (1m)
2026-09-15 19:56:35	10.0.0.21	DDoS	UDP Flood	75.3%	High	Time Ban
2026-09-15 19:56:37	10.0.0.22	DDoS	UDP Flood	75.9%	High	Time Ban (1m)
2026-09-15 19:56:37	10.0.0.22	DDoS	UDP Flood	75.9%	High	Time Ban
2026-09-15 19:57:23	10.0.0.8	DDoS	UDP Flood	91.8%	High	Time Ban (1m)
2026-09-15 19:57:34	10.0.0.20	DDoS	UDP Flood	80.2%	High	Time Ban (2m)
2026-09-15 19:57:35	10.0.0.21	DDoS	UDP Flood	77.4%	High	Time Ban (2m)
2026-09-15 19:57:37	10.0.0.22	DDoS	UDP Flood	77.1%	Critical	Time Ban (2m)
2026-09-15 19:58:24	10.0.0.8	DDoS	UDP Flood	91.8%	High	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.8	UDP Flood	0.5804	91.8%	High	Phase 2	1	1m 1s	2026-09-15 19:58:24	Ban Expired
10.0.0.22	UDP Flood	0.6247	77.1%	Critical	Phase 2	2	1m 11s	2026-09-15 19:57:37	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6392	77.4%	High	Phase 2	2	1m 11s	2026-09-15 19:57:35	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6398	80.2%	High	Phase 2	2	1m 11s	2026-09-15 19:57:34	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6262	75.9%	High	Phase 2	1	11s	2026-09-15 19:56:37	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6451	75.3%	High	Phase 2	1	11s	2026-09-15 19:56:35	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	75.6%	High	Phase 2	1	11s	2026-09-15 19:56:34	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE